
Revision 3: Calliophis CK — the abundant "cytotoxin" is maticotoxin, and it was tested

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Creatine Kinase Elevation in *Calliophis* Envenoming

Two candidate mechanisms, and how to tell them apart. — Revision 3, 7 October 2026

What changed since Revision 2. Revision 2 noted that the "cytotoxins" in this venom are annotations by sequence homology rather than demonstrated activities. Revision 3 adds the sharper version.

The abundant "cytotoxin-like" fraction is maticotoxin. Tasoulis, Pukala and Isbister, correcting another paper's abundance arithmetic, state that in *C. bivirgata flaviceps* **one toxin — maticotoxin — makes up 23% of the whole venom.**

Maticotoxin is the toxin Takasaki et al. isolated and characterised in 1991. They tested its activity, and per Dashevsky and Fry **it is not cytotoxic**. Maticotoxin A was noted at the time to resemble cytotoxins from other elapids — a sequence observation, which is where the label came from.

So the most abundant single toxin in this venom was isolated, assayed, and found not to be a cytotoxin — and has been called one for thirty-five years anyway.

Two effects: it strengthens the critique of Mechanism A, which loses its cytotoxin component entirely; and it *partly reverses* Revision 2's abundance worry — maticotoxins and δ -elapitoxins are distinct subclades, so this fraction is not calliotoxin in disguise. Calliotoxin is probably a genuinely minor protein component, and Mechanism B weakens accordingly. §3.3 is also new: how thin the direct cytotoxicity evidence actually is.

1. Summary

(A) Direct myotoxicity, principally from PLA₂. Reasonably supported. PLA₂ dominates the proteome by mass, whole venom is demonstrably myotoxic in mice, and PLA₂ myotoxicity is well characterised across elapids.

(B) Channel-mediated myotoxicity from calliotoxin at Na_v1.4. Speculative, and weakened by the abundance finding above.

They are not mutually exclusive, and they predict different EMG findings.

2. The Clinical Record

Source	Species	n	Findings
Takasaki 1991 (footnote)	<i>M. bivirgata</i>	1	Wet bite, researcher. Pain and swelling; recurrent pain ~1 month. Minor symptoms only.
Tansuwannarat 2026	<i>C. b. flaviceps</i>	7	Swelling 4/7, pain 2/7. One severe pain requiring IV analgesia. One anaphylaxis. No neurotoxicity.
	<i>M.</i>		Fatal, child, ~2 h after bite on thumb web. No clinical description

Harrison 1957 *bivirgata* 1 survives.

Handbook §37.5.6	<i>Calliophis</i> spp.	—	Local effects self-limiting within days. Elevated CK in some patients, no significant complications.
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Across 52 Asian coral snake bites in Thailand: no ptosis, bulbar weakness, respiratory compromise, paralysis or death; no ventilation or antivenom. The authors state CK **was not systematically measured** and subclinical myotoxicity could not be ruled out — so the largest series available is silent on the question §37.5.6 answers.

The dose question is settled: Takasaki's was a wet bite from a snake that can yield ~150 mg in one milking.

3. Mechanism A: Direct Myotoxicity

3.1 What the venom contains

Tan et al. 2016 (*C. bivirgata flaviceps*): 23 proteins across 11 families, predominated by PLA₂ and cytotoxin, **no detectable post-synaptic neurotoxin**. Moderate cytotoxicity against fibroblasts and myotoxicity in mice. ~41% PLA₂, ~23% cytotoxin-like.

Tan et al. 2019 (*C. intestinalis*, Ismail a co-author): PLA₂ 43.4%, 3FTx 20.1%. In vivo **myotoxic but non-lethal** to mice, frogs and geckos even at 5–10 µg/g.

3.2 The “cytotoxin” problem

The word *cytotoxin* does a great deal of work in this literature, and it is not earned.

The annotations are by homology. No purified *Calliophis* “cytotoxin” has been shown to lyse cells.

Yang et al. made this criticism in 2016. Earlier studies concluding the venom's 3FTx were exclusively cytotoxic reflected a very narrow scope of assays, with cytotoxicity *assumed* from partial-sequence similarity rather than established by activity characterisation. Their own work then found a 3FTx doing something else entirely.

Dashevsky and Fry went further in 2021. Although cytotoxins are close neighbours of the clades containing the δ-elapitoxins and maticotoxins, the studies describing both noted that **they are not themselves cytotoxic**. The containing clade sits in a polytomy that also includes a large group of *Micrurus* toxins never shown to be cytotoxic.

And the loop closes. Tasoulis et al. state that maticotoxin alone makes up 23% of the whole venom — identifying the “cytotoxin-like” fraction as the toxin Takasaki characterised and found not to be cytotoxic.

Even the authors who use the label hedge it. Palasuberniam et al. propose these as **Y-type CTX-like proteins**, preserving a *less toxic* form of elapid CTX-like protein.

3.3 What direct evidence of cytotoxicity actually exists

One assay. Tan et al. 2016 measured *crude venom* against CRL-2648 fibroblasts: IC₅₀ 62.14 ± 0.87 µg/mL, described as moderate. Two limitations:

- **It is crude venom** — it cannot attribute the effect to any component. PLA₂ at 41% would produce membrane damage unaided.
- **The figure is high** for a venom alleged to be cytotoxin-rich. Genuinely cytotoxin-dominated elapid venoms are considerably more potent in this kind of assay.

Nothing else exists. The nanofractionation platforms developed since 2021 profile elapid cytotoxicity across human cell lines *after* separating components — precisely the method that would settle this. Applied to *Naja*, *Bungarus* and others. *Calliophis* has not been run.

3.4 Consequences

Mechanism A loses its cytotoxin component entirely. What remains is PLA₂ at ~41%. The whole venom's cytotoxicity

and myotoxicity are real measured findings, but they were assigned to cytotoxin or cytotoxin–PLA₂ synergy *presumptively*, on the same contested annotation. PLA₂ alone accounts for both.

The abundance question partly resolves in favour of §5's original claim. Calliotoxin is probably a genuinely minor protein component, and **Mechanism B weakens accordingly.**

What replaces the worry is a better question. A toxin at 23% of this venom, isolated thirty-five years ago, with **no established function** — not cytotoxic, not demonstrated at any channel.

3.5 The recommendation already exists

From Tan et al. 2019: serum creatine kinase should be monitored serially for potential complication of myotoxicity. This is the Handbook's observation, in print, with a mechanism attached, from the same clinician.

3.6 How PLA₂ myotoxicity works

Myotoxic PLA₂s bind the sarcolemma and hydrolyse its phospholipids; the membrane loses integrity, calcium enters, cytosolic contents including CK escape. A direct coral-snake precedent: *Micrurus nigrocinctus* venom acts primarily on the sarcolemma, producing wedge-shaped “delta” lesions within 15 minutes with rapid loss of muscle CK and creatine and calcium influx.

3.7 Why this explanation is still preferred

- PLA₂ is dominant by mass, with an independently established myotoxic mechanism.
- Myotoxicity demonstrated in mice for both species.
- No assumptions about dose, kinetics or channel behaviour.
- Explains local, self-limiting CK elevation and the absence of neurotoxicity.

It remains parsimonious, though on a narrower base: PLA₂ alone, rather than PLA₂ plus a cytotoxin fraction that turns out not to be cytotoxic.

4. Mechanism B: Na_v1.4 and Calliotoxin

4.1 The channel

Na_v1.4 is the voltage-gated sodium channel of adult skeletal muscle, encoded by *SCN4A* (17q23.3) — the predominant isoform in adult skeletal muscle, generating the inward current behind the action-potential upstroke.

The α-subunit folds into four homologous domains around a central pore, each with six transmembrane segments: S1–S4 form the voltage sensor (S4 carries positive charges that move outward on depolarisation); S5–S6 line the pore.

Three states, two gates: closed at ~–80 to –90 mV; open on depolarisation (sub-millisecond); then fast-inactivated, when the DIII–DIV linker swings across the inner mouth of the pore and plugs it. Fast inactivation is separate from closing the activation gate and can be disrupted independently. That is what calliotoxin does.

4.2 What calliotoxin does

Parameter	Control	+ Calliotoxin	Effect
V _½ activation	–35.0 ± 0.1 mV	–38.1 ± 0.3 mV	Hyperpolarising shift
V _½ fast inactivation	–66.1 ± 0.5 mV	–61.6 ± 0.7 mV	Depolarising shift
Inactivation τ (–20 mV)	0.24 ± 0.02 ms	5.22 ± 0.9 ms	~22× slowing
Persistent current	–6 ± 4 pA	–536 ± 33 pA	~90× increase

Peak ramp current	-143 ± 12 pA	-728 ± 52 pA	~5× increase
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200 nM purified native toxin, HEK-293 cells expressing **human** Na_v1.4. All effects abolished by tetrodotoxin; unaffected by nicotinic, muscarinic and α₁ antagonists. In intact muscle: rising contractions, fasciculations, abolished nerve-mediated twitches. Yang et al. report no taxon-specific effect — so the mild clinical picture cannot be attributed to channel-level resistance.

4.3 The analogy to human *SCN4A* disease

Inference from here on.

Inherited gain-of-function *SCN4A* mutations cause sodium-channel myotonia, paramyotonia congenita, and hyperkalaemic/normokalaemic periodic paralysis. What impaired fast inactivation does to a living human:

- **Myotonia** — stiffness, delayed relaxation.
- **Episodic weakness** when depolarisation inactivates channels secondarily.
- **Myotonic discharges on EMG.**
- **CK elevation.** In 56 paramyotonia congenita patients (49 with *SCN4A* mutations), about half had mild weakness and **most had increased serum CK**. Values are modest: 848 U/L (T704M), 279–771 U/L (paediatric), 276 IU/L. Recurrent exertional rhabdomyolysis also reported in a paramyotonia family with *SCN4A* variants.

This establishes plausibility, not causation.

4.4 Precedents from other Na_v activators

Scorpion envenoming. A documented rhabdomyolysis-and-myocarditis case: CK 2,129 U/L, CK-MB 43.24 ng/mL, troponin 22.59 ng/mL. Cardiac markers show this was not purely skeletal, and scorpion envenoming involves massive catecholamine release — the CK cannot be attributed cleanly to direct channel action.

Ciguatera. Ciguatoxins bind site 5 on voltage-sensitive channels in all excitable tissues; P-CTX-1 acts across multiple Na_v isoforms. Symptoms include myalgia, arthralgia, paraesthesiae, dysaesthesiae and heightened nociception, some persisting months to years. The more interesting comparison for the Takasaki case.

5. On the “Low Abundance” Claim

Handbook §37.2.1.4 cites Yang et al. 2016. **The citation appears misattributed** — they did not quantify callitoxin's share, and their argument runs the other way.

Revision 2 doubted the claim; Revision 3 largely restores it. The abundant fraction is maticotoxin, and maticotoxins and δ-elapitoxins are distinct subclades — so it is not callitoxin in disguise.

Conclusion: callitoxin probably is a minor protein component, the Handbook's claim is probably right, and only its citation is wrong. Mechanism B weakens correspondingly.

The transcript–protein divergence remains striking and unexplained — ~40% of toxin *transcripts* versus a minor share of secreted *protein*. Post-transcriptional regulation, differential secretion, or poor peptide recovery are all possible. None investigated.

Still worth doing: re-analysis of the existing proteomic spectra against the 2021 transcriptome, which did not exist when the proteome was annotated. It would put a number on callitoxin directly rather than by subtraction. Computational reanalysis of published data, no new experiment.

6. Distinguishing the Two Mechanisms

Finding	A (direct, PLA ₂)	B (callitoxin/Na _v 1.4)
Myotonic discharges		Present — impaired fast inactivation is the

on EMG	Absent	classic cause
Fasciculations near bite site	Not expected	Expected
Stiffness / delayed relaxation	Not expected	Expected
CK time course	Peaks 24–48 h, falls over days	Tracks toxin presence; earlier peak
Spatial distribution	Confined to envenomed tissue	Potentially wider
Na ⁺ -channel blockade	No effect	Mexiletine predicted to help
Histology	Delta lesions, sarcolemmal disruption	Fibre damage without primary lysis

The single most informative test is needle EMG near the bite site in the acute phase. Myotonic discharges would be strong evidence that callitoxin acts on human Na_v1.4 in vivo — never sought, as far as I can determine. Their absence, with raised CK, would point firmly to Mechanism A.

Other predictions: serial CK with CK-MB; CK elevation commoner than clinical findings; myoglobinuria absent or trivial; **species comparison** (*C. intestinalis* vs *C. bivirgatus* — requires only existing poison-centre records); proteome reanalysis.

The maticotoxin gap

Separately from the CK question, the most conspicuous hole in this venom's characterisation is **a toxin making up roughly a quarter of it whose function is unknown**. Maticotoxin was isolated in 1991, is not cytotoxic, and has not been tested at any ion channel or receptor. Only a partial sequence for maticotoxin A was published; gross amino acid compositions were given for the others, which Dashevsky and Fry note may never be matchable to specific sequences.

Three things worth doing, none requiring a patient:

- **Run it through a modern functional panel** — nicotinic and muscarinic receptors, Na_v subtypes, K_v, Ca_v. Callitoxin was found because somebody ran an assay nobody had thought to run; maticotoxin has had no equivalent attempt.
- **Test purified maticotoxin for cytotoxicity properly**, by nanofractionation against human cell lines. Settles a thirty-five-year-old label either way.
- **Recover full sequences** from the 2021 transcriptome and match them to the 1991 fractions.

If maticotoxin shares callitoxin's target, Mechanism B returns with a quarter of the venom behind it. If it is genuinely cytotoxic, Mechanism A is vindicated on evidence rather than assumption. Either result beats the present position.

Study design. Prospective cohort at a Malaysian or Thai centre: serial CK, CK-MB, renal function, potassium, urine myoglobin, needle EMG where feasible, stratified by species. Given seven confirmed *C. bivirgatus* bites in ten years in Thailand, this needs multi-centre collaboration. The retrospective comparison could be done now.

7. What Should Be Retired

“The majority of human *C. bivirgatus* envenomations were lethal.” Asserted on the basis of a single 1957 case with no surviving clinical description. Against it stand eight or more documented bites — including one large wet bite — with minor or local effects, no systemic neurotoxicity, no deaths.

That a bite is a medical emergency. The evidence supports assessment and short observation, which is what the Thai authors recommend as a precaution rather than on the basis of anything observed.

8. Caveats

- §37.5.6 is a personal communication: no case count, values, species breakdown, or treatment of confounders.
- CK is non-specific.
- Mechanism B remains unevidenced in humans.
- Calliotoxin's subtype selectivity is unknown — only Na_v1.4 was tested.
- The fatal case remains unexplained and probably always will.
- Species pooling and venom variation both limit interpretation.
- **The cytotoxin critique has limits.** §3.2 establishes that maticotoxin was characterised and found not cytotoxic, and that the label persisted anyway. It does not establish that maticotoxin has *no* membrane activity under conditions nobody has tested, nor that the minor 3FTx components are innocent. Crude venom cytotoxicity and myotoxicity are real measured findings; what is unresolved is which components produce them.
- **The maticotoxin identification rests on one source.** Tasoulis et al. state the 23% figure in a commentary on another paper, not a primary analysis of this venom. Consistent with the published proteome, but worth verifying against source data.

9. Conclusion

CK elevation in *Calliophis* bite patients is most likely direct myotoxicity, driven principally by the PLA₂ fraction that dominates these venoms — an explanation already proposed in print, by the same clinician whose observation the Handbook records.

Whether calliotoxin also contributes through Na_v1.4 is open and plausible on mechanism, **but probably minor**: the abundant three-finger toxin in this venom is maticotoxin, not calliotoxin. It remains worth excluding, because it is cheap to do and because a positive finding would be the first demonstration that this unusual toxin does anything at all in a person.

The larger prize is elsewhere. A toxin constituting roughly a quarter of this venom has no known function. Calliotoxin was found because somebody ran an assay nobody had thought to run. Maticotoxin has never had that attempt, and the label it carries is very likely the reason: **a protein already filed under “cytotoxin” does not look like an open question.**

Two broader lessons emerge, and they are versions of the same failure.

A reputation outran its evidence. The danger attributed to this snake rests on one undocumented death from 1957, amplified through citation into assertions about lethality the clinical record contradicts.

An annotation outran its evidence. “Cytotoxin” entered this literature as a sequence-similarity label and has been read ever since as a demonstrated function — to the point of being offered as the explanation for clinical findings. Yang et al. flagged the problem in 2016; Dashevsky and Fry restated it in 2021. The label persists regardless, attached to a protein whose own 1991 characterisation contradicted it.

In both cases a provisional claim hardened through repetition into a fact nobody had established. Anyone working on this venom should treat the descriptive vocabulary — “cytotoxin”, “medically important”, “potentially lethal” — as inherited assumptions rather than findings, and check what each one actually rests on.

Sources

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Revision 3, 7 October 2026. §3 reports published findings, with §§3.2–3.3 summarising published critiques and the absence of direct evidence. §§4, 6 and 7 contain inference and hypothesis, flagged in place.